

Entering the Car Docker

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1. Course Content

- This lesson is only for users with **Raspberry Pi 5** and **Jetson-Nano** motherboards.

[!NOTE]

- Because the Raspberry Pi 5 and Jetson Nano system versions cannot directly install the `ros2 humble` environment, the `ros2 humble` environment is placed in `Docker`, and the host machine does not have a ROS2 environment.
- After entering Docker with the Jetson Nano version, you need to open an rviz2 or rqt window before running the AI large-scale multimodal vision tutorial.

2. Basic Operations

2.1 Launching the m3pro Container

- If you wish to run the example programs featured in this tutorial, you must first launch the `m3pro` Docker container.

[!WARNING]

- **Important:** Ensure that you launch the `m3pro` Docker container *within* the VNC graphical interface; otherwise, visualization windows (such as RViz) may fail to display correctly.
- Regarding external peripherals—such as game controllers, AI voice modules, and other hardware devices—the Docker container will only load devices that were present at the moment the container was launched. If a hardware device is plugged in *after* the container has already started, it will not be automatically synchronized in real-time within the container. To recognize newly connected devices, you must stop and then restart the container.

```
bringup_m3pro
```

The message `container rosmaster-m3pro started` indicates that the launch was successful.

```
pi@raspberrypi: ~
pi@raspberrypi: ~ 80x24
[System Information]
-----
IP_Address_1: 192.168.12.51
IP_Address_2: 172.19.0.1
MACHINE: RaspberryPi5 | ROS_DISTRO: humble
ROBOT_TYPE: ROSMASTER-M3Pro | CAMERA_TYPE: dabai_dcw2 | RADAR: Tmini-plus*2
-----
pi@raspberrypi:~ $ bringup_m3pro
access control disabled, clients can connect from any host
[+] start 1/1
✓ Container rosmaster-m3pro Started 0.2s
pi@raspberrypi:~ $
```

2.2 Accessing the m3pro Container Terminal

- The commands featured in the subsequent tutorials must be executed *inside* the container. To open the container terminal, use the following command:

```
exec_m3pro
```

```
root@raspberrypi: ~
root@raspberrypi: ~ 80x24
pi@raspberrypi:~ $ exec_m3pro
MY_DOMAIN_ID: 30
root@raspberrypi:~#
```

- To exit the container terminal, press `Ctrl + D`.

```
MY_DOMAIN_ID: 30
root@raspberrypi:~#
exit
pi@raspberrypi:~ $
```

[!IMPORTANT]

Important

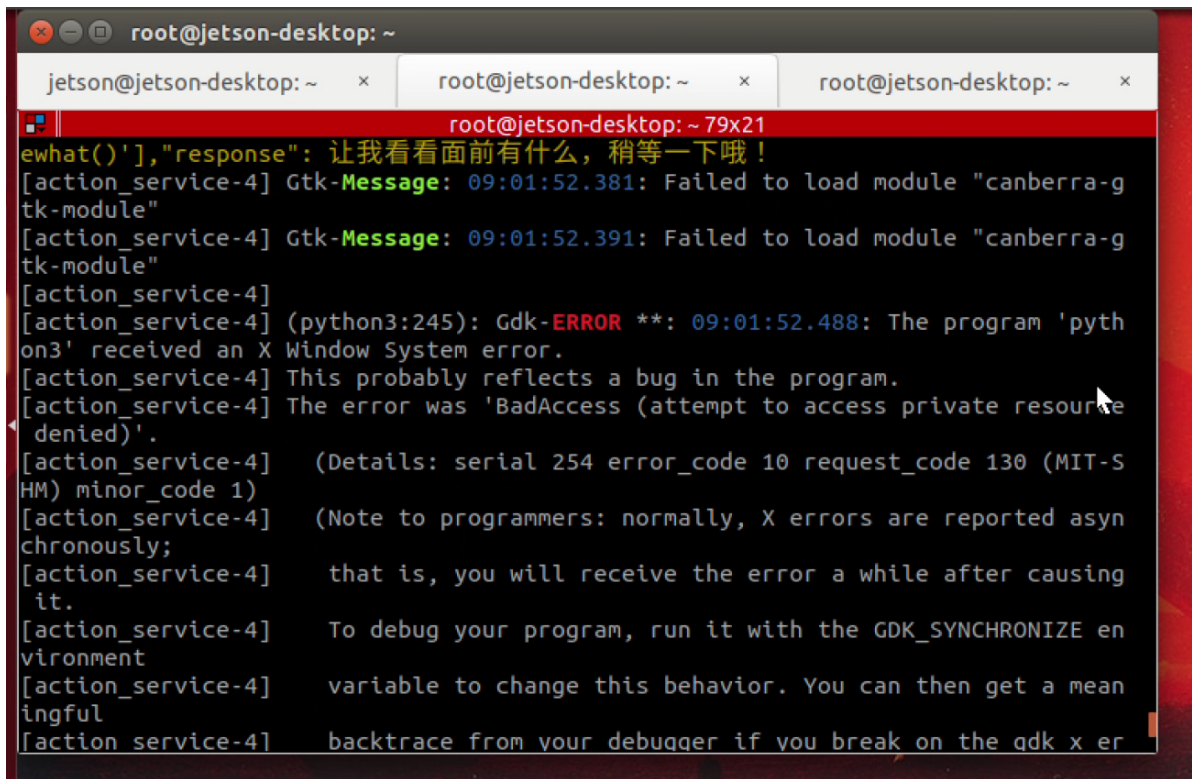
- Unless explicitly stated otherwise, users of the Raspberry Pi and Jetson Nano platforms should assume that all subsequent program execution commands are to be performed *within* the **m3pro** container. You must first launch the container and then enter the commands within the container terminal. This requirement will not be reiterated in subsequent lessons. ### 2.3 Shutting Down the m3pro Container
- If you genuinely no longer require the m3pro container and need to shut it down manually (typically, manual shutdown is not necessary):

```
shut_m3pro
```

```
pi@raspberrypi: ~
pi@raspberrypi: ~ 80x24
pi@raspberrypi:~ $ shut_m3pro
[+] stop 1/1
pi@raspberrypi:~ $ ter-m3pro Stopping 10.5s
```

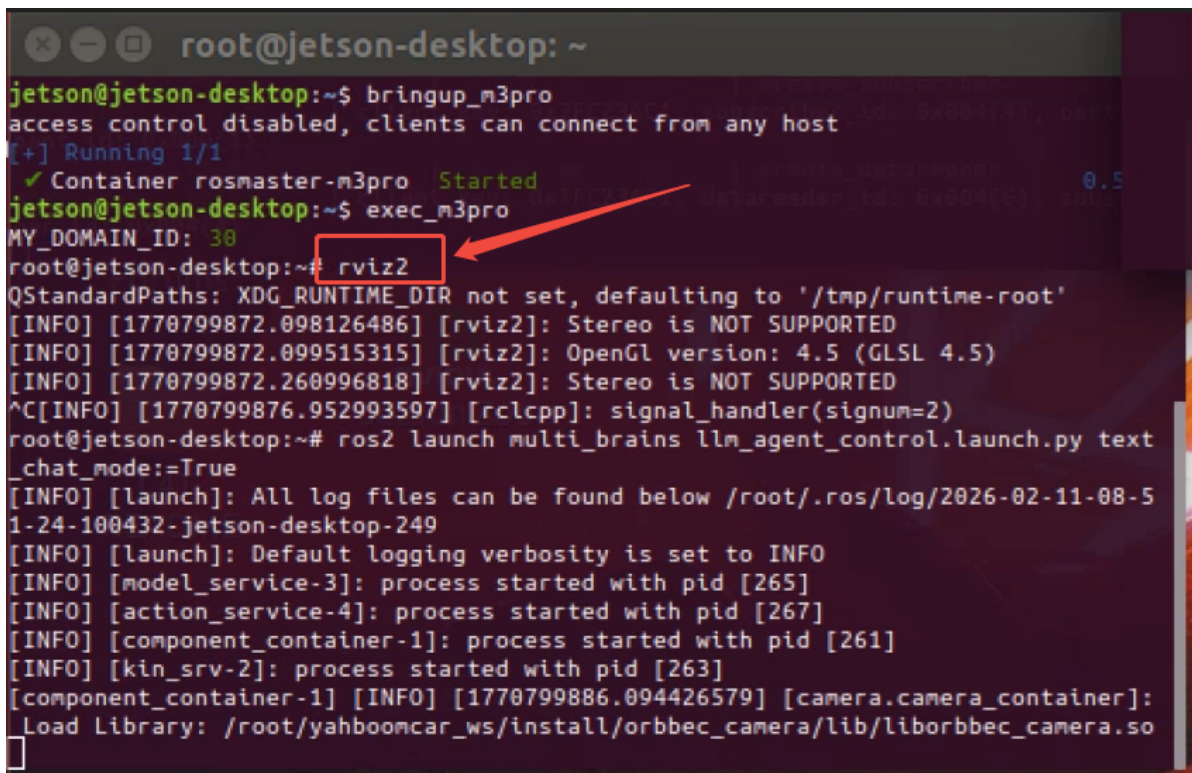
2.4 Issues Encountered After Entering the Docker Container on Jetson Nano Versions of M3Pro, A1, M1, and M3 Robots

- On Jetson Nano versions, attempting to run the AI Large Model Vision examples immediately after entering the Docker container results in a "Permission Denied" error, as shown in the figure below.



```
root@jetson-desktop: ~
jetson@jetson-desktop: ~ x root@jetson-desktop: ~ x root@jetson-desktop: ~ x
root@jetson-desktop: ~ 79x21
ewhat()'], "response": 让我看看面前有什么，稍等一下哦！
[action_service-4] Gtk-Message: 09:01:52.381: Failed to load module "canberra-gtk-module"
[action_service-4] Gtk-Message: 09:01:52.391: Failed to load module "canberra-gtk-module"
[action_service-4]
[action_service-4] (python3:245): Gdk-ERROR **: 09:01:52.488: The program 'python3' received an X Window System error.
[action_service-4] This probably reflects a bug in the program.
[action_service-4] The error was 'BadAccess (attempt to access private resource denied)'.
[action_service-4] (Details: serial 254 error_code 10 request_code 130 (MIT-SHM) minor_code 1)
[action_service-4] (Note to programmers: normally, X errors are reported asynchronously;
[action_service-4] that is, you will receive the error a while after causing it.
[action_service-4] To debug your program, run it with the GDK_SYNCHRONIZE environment
[action_service-4] variable to change this behavior. You can then get a meaningful
[action_service-4] backtrace from your debugger if you break on the gdk_x_error function.
```

- Solution: First, open a window within the Docker container using commands such as `rviz2` or `rqt`. After opening and then closing this window, you may proceed to run the AI Large Model Vision examples, as illustrated in the figure below.



```
root@jetson-desktop: ~
jetson@jetson-desktop:~$ bringup_m3pro
access control disabled, clients can connect from any host
[+] Running 1/1
  ✓ Container rosmaster-m3pro Started
jetson@jetson-desktop:~$ exec_m3pro
MY_DOMAIN_ID: 30
root@jetson-desktop:~# rviz2
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
[INFO] [1770799872.098126486] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1770799872.099515315] [rviz2]: OpenGL version: 4.5 (GLSL 4.5)
[INFO] [1770799872.260996818] [rviz2]: Stereo is NOT SUPPORTED
^C[INFO] [1770799876.952993597] [rclcpp]: signal_handler(signum=2)
root@jetson-desktop:~# ros2 launch multi_brains_llm_agent_control.launch.py text_chat_mode:=True
[INFO] [launch]: All log files can be found below /root/.ros/log/2026-02-11-08-51-24-100432-jetson-desktop-249
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [model_service-3]: process started with pid [265]
[INFO] [action_service-4]: process started with pid [267]
[INFO] [component_container-1]: process started with pid [261]
[INFO] [kin_srv-2]: process started with pid [263]
[component_container-1] [INFO] [1770799886.094426579] [camera.camera_container]: Load Library: /root/yahboomcar_ws/install/orbbec_camera/lib/liborbbec_camera.so
```

